

Understanding the Effect of Shortening Compulsory Military Service

Evidence From Taiwan

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1. Introduction
2. Literature Review
3. Data
4. Estimation
5. Extension
6. Conclusion

Compulsory Military Service in Taiwan

- Length overall change: 2 yr $\rightarrow$ 1 yr $\rightarrow$ 4 mon $\rightarrow$ 1 yr Appendix
- Reform in 2013: Men born in or after 1994 are required to serve only 4 mon.
- Main question: How this policy affects one's latter outcomes?
 - Especially job search and wage earnings

1. Introduction
2. Literature Review
3. Data
4. Estimation
5. Extension
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Literature Review

- Bauer, Bender, Paloyo, and Schmidt (2009)

When considering selection effects, the earnings advantage of those who served in the army disappear.

- Chuang (2019)

Accumulated 1.56 months of experience doesn't have effect on their first job income.

- Mouganie (2020)

Significant increase in years of education but has no effect on employment and wages at the ages of 30–36.

- Milovanska-Farrington (2024)

Individuals started to have their first child earlier given that they decide to have children after the suspension of the draft.

Hypothesis

- Suppose two identical individuals make similar lifetime decisions
 - college $\rightarrow$ military service $\rightarrow$ first job
- The only difference is their birth year (younger vs. older cohort)
- Military service $\downarrow \rightarrow$ earlier entry into the labor market $\rightarrow$ wage at age τ $\uparrow$
 - e.g. Income at age 25 should be discontinuous around 1994.

1. Introduction
2. Literature Review
- 3. Data**
4. Estimation
5. Extension
6. Conclusion

Data

- Survey on the Employment Situation of Young Labor (15-29)
- Survey Year: 101, 103, 105, 107, 108, 109, 111
 - Repeated Cross-Sectional Data
- We used the following variables:
 - ① Dependent Variables: find_job_length, current_wage, first_job_wage
 - ② Independent Variables: age, birth_yr
 - ③ Control Variables: gender, first_job_age
 - ④ Others Variables: first_job_yr, first_job_month

Define Variables

- $\text{first_job_length} = \text{first_job_yr} + \text{first_job_month}/12$
- $\text{relative_birth_yr} = \text{birth_yr} - 1994$
- educ
 - ① junior high school or below = 1
 - ② high school or vocational high school = 2
 - ③ junior college = 3
 - ④ bachelor = 4
 - ⑤ master = 5
 - ⑥ Ph.D = 6

Sample Selection

- Age: 18-29
- We can only assume each male is required for military service.
- Sample Size: 12,434

Data Limitations

- The data only reveals the participants' birth year, but not birth month
- Survey data is not available for every year
- The overall data quality is not great that we cannot run a RD design

Identification Assumption

- Continuous Conditional Expectation Function (CEF)
 - But we have discrete Running Variable
- Local Randomization
 - But we have insufficient data points given arbitrary bandwidth
 - This may lead to a biased estimator
- Maybe administrative data is better for this research question

1. Introduction
2. Literature Review
3. Data
- 4. Estimation**
5. Extension
6. Conclusion

Empirical Strategy: Canonical Setup

- Suppose we have the administrative data

$$y_{it} = \beta_0 + \beta_1 \text{reform}_i + \beta_2 (R_i - 1994) + X' \eta + \varepsilon_{it}$$

- y_{it} : Wage at the given year
- R_i : running variable: birth year, with 1994 as the cutoff
 - $\text{reform}_i = \mathbf{1}\{R_i \geq 1994\}$
 - Ideally, we should have birth year and month!
- X : bachelor_{it}
- β_1 : the coefficient of interest

Empirical Strategy: Survey Data Setup

- But now we only have survey year data!
- Consider the following setup

$$y_{it} = \beta_0 + \beta_1 \text{reform}_i + \beta_2 (R_i - 1994) + X' \gamma + \xi_{it}$$

- y_{it} : wage at the given year
- $\text{reform}_i = \mathbf{1}\{R_i \geq 1994\}$
- X : bachelor_{it} , survey year fixed effects, age fixed effects

Empirical Strategy: Problem

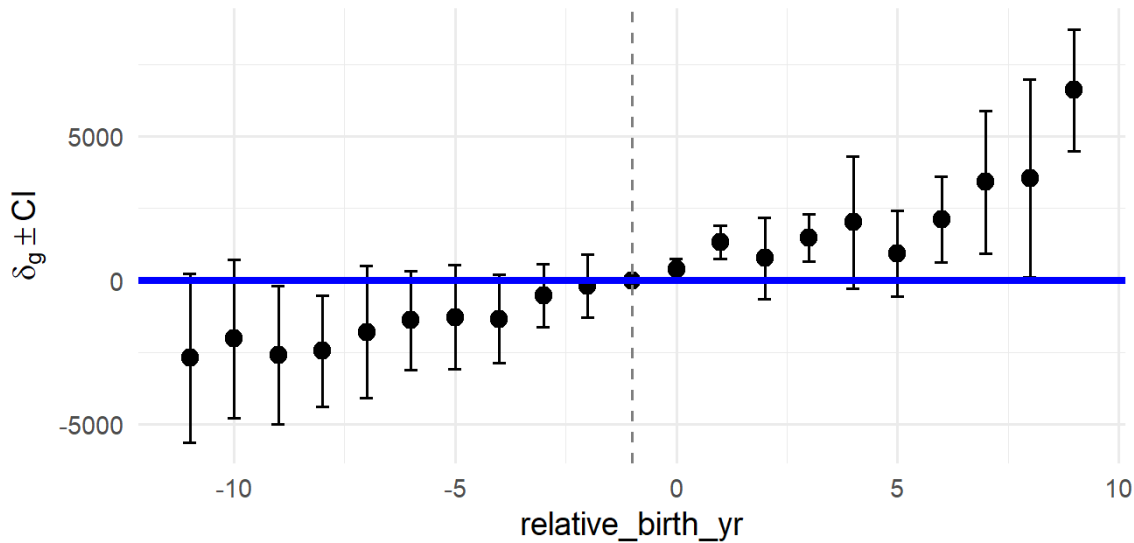
- $\text{birth_yr} + \text{age} = \text{survey_yr}$
- When including both survey yr and age fixed effects, it would occur a co-linearity problem.
- When only considering survey yr fixed effects, age would confound with wage.
- When only considering age fixed effects, survey year would confound with wage.

Event-Study Regression

- Let's consider event-study approach

$$wage_{it} = \sum_{g \neq -1} \delta_g \cdot \mathbf{1}\{R_i = g\} + \beta \cdot bachelor_{it} + \lambda_a + \gamma_t + v_{it}$$

- $wage_{it}$: monthly income
- R_i : relative birth year; $g \in \{-11, -10, \dots, 10\} \setminus \{-1\}$
- $bachelor_{it}$: attending college or not
- FE: λ_a (age FE); γ_t (survey yr FE)
- We expect $\delta_g > 0$ as $R_i \geq 0$



Event-Study Regression (Conti.)

- An upward pattern is observed, as expected, but it is not robust
- Bachelor degree matters
- Detailed regression table [Appendix](#)

1. Introduction
2. Literature Review
3. Data
4. Estimation
- 5. Extension**
6. Conclusion

Future Research Plan

- Reform in 2024: how to conduct counterfactual analysis?
- Limitation of reduced-form approach
- Recall our story, we assume everyone experiences a similar trajectory
- We allow some heterogeneity and dynamic decisions
- Modelling job search and career choice (In progress)

Dynamic Discrete Choice Model

- Basic human capital model
- Following Keane and Wolpin (1997) with mild adjustment
- Taiwanese male agent faces a finite horizon problem beginning with age 18 and ending at age $T = 36$.

State Variables

- s : years of schooling
- x_1, x_2 : occupation-specific work experience (full-time or part-time)
- z : years of military service
- $\xi(a)$: idiosyncratic shock for choice a , $a = 1, 2, \dots, 5$

Control Variables

Define a Choice set $A = \{1, 2, 3, 4, 5\}$:

- $a = 1$: work in a blue-collar occupation
- $a = 2$: work in a white-collar occupation
- $a = 3$: mandated to serve in the military or volunteer to work in the military
- $a = 4$: tertiary education
- $a = 5$: remain at home

Deep Parameters

- θ_0^a , $a \in \{1, 2, 4\}$: return to education
- θ_1^a , $a \in \{1, 2\}$: return to work experience (occupation-specific)
- η : depreciation rate as a soldier
- $\underline{e}$: the obligation and basic earnings as a soldier
- c_1 : cost of attending college
- c_2 : cost of attending graduate school or pursue ph.D
- β : discounted factor (calibrated)

Flow Utility Function

- Ind. i in each period t

$$U(\mathbf{X}_{it}, \boldsymbol{\xi}_{it}, a_{it}) = \begin{cases} \log(\theta_0^1 s_{it} + \theta_1^1 x_{1,it}) + \xi_{it}(1) & \text{if } a_{it} = 1 \\ \log(\theta_0^2 s_{it} + \theta_1^2 x_{2,it}) + \xi_{it}(2) & \text{if } a_{it} = 2 \\ \log(\underline{e} + \eta z_{it}) + \xi_{it}(3) & \text{if } a_{it} = 3 \\ \log(\theta_0^4 s_{it} - c_1 \mathbf{1}_{12 \leq s_{it} \leq 16} - c_2 \mathbf{1}_{s_{it} \geq 16}) + \xi_{it}(4) & \text{if } a_{it} = 4 \\ \xi_{it}(5) & \text{if } a_{it} = 5, \end{cases} \quad (1)$$

Rust Assumptions

- 1 Additive separability

$$U(\mathbf{X}_{it}, \boldsymbol{\xi}_{it}, a_{it}) = u(\mathbf{X}_{it}, \boldsymbol{\xi}_{it}, a_{it}) + \xi(a_{it})$$

- 2 I.I.D. TIEV

$$\xi_{it}(a_{it}) \stackrel{iid}{\sim} G(\xi_{it})$$

$$\mathbf{X}_{it} \perp \xi_{it}$$

Law of Motions

- Schooling

$$s'(t+1) = s(t) + \mathbb{1}\{a_t = 4\}, \quad s', s \leq \bar{S}, \quad 18 \leq t \leq 36,$$

- Work experience

$$x'_m(t+1) = x_m(t) + \mathbb{1}\{a_t = m\}, \quad m = 1, 2, \quad 18 \leq t \leq 36,$$

- Military Service

$$z'(t+1) = z(t) + \mathbb{1}\{a_t = 3\}, \quad 18 \leq t \leq 36.$$

Lifetime Maximization Problem

$$\begin{aligned} V(\mathbf{X}_t) &= \max_{a_t \in A} \mathbb{E} \left(\sum_{\tau=0}^{T-t} \beta^\tau \left\{ u(\mathbf{X}_{t+\tau}, a_{t+\tau}) + \xi_{t+\tau}(a_{t+\tau}) \right\} \mid \mathbf{X}_t, \xi_t, a_t \right) \\ &= \max_{a_t \in A} \left[u(\mathbf{X}_t, a_t) + \xi_t(a_t) + \beta \iint V(\mathbf{X}_{t+1}, \xi_{t+1}) dG(\xi_{it}) dF(\mathbf{X}_{t+1} \mid \mathbf{X}_t, a_t) \right]. \end{aligned} \tag{2}$$

Conditional Choice Probability

- CCP

$$\begin{aligned}\Pr(a_t = j \mid \mathbf{X}_t) &= \Pr(v_j + \xi(j) > v_k + \xi(k), k \neq j \mid \mathbf{X}_t) \\ &= \frac{\exp(v_j)}{\sum_{a \in A} \exp(v_a)}\end{aligned}\quad (3)$$

- Let $EV(\mathbf{X}) = \mathbb{E}_\xi [V(\mathbf{X}_t, \xi_t)]$, one can show that

$$\begin{aligned}EV(\mathbf{X}_t) &= \log \left\{ \sum_{a_t} \exp \left(u(\mathbf{X}_t, a_t) + \beta \mathbb{E}_{X_{t+1}} [EV(\mathbf{X}_{t+1}) | \mathbf{X}_t, a_t] \right) \right\} + \gamma \\ &= \log \left\{ \sum_{a_t} \exp(v(\mathbf{X}_t, a_t)) \right\} + \gamma,\end{aligned}\quad (4)$$

- $v(\cdot)$: choice-specific conditional value function (w/o ξ_t)
- γ is the Euler-Mascheroni constant.

Identification

- Our goal is to recover $\Theta = \left\{ \theta_0^1, \theta_0^2, \theta_0^4, \theta_1^1, \theta_1^2, \underline{e}, \eta, c_1, c_2 \right\}$.
- Likelihood Function

$$\mathcal{L}(\Theta \mid \mathbf{X}_{it}, a_{it}) = \prod_{n=1}^N \prod_{t=18}^{36} \prod_{j=1}^5 \Pr(a_{it} = j \mid \mathbf{X}_{it})^{\mathbb{1}\{a_{it}=j\}} \quad (5)$$

Estimation

- Nested Fixed Point Algorithm (NFXP)
 - ① Initial guess of a parameter set $\hat{\Theta}_0$.
 - ② Inner loop: solve integrated value function (integrate over ξ) (4) via value function iteration.
 - ③ Construct CCP (3), where choice specific value function $v(\cdot)$ captures the transition of state $\mathbf{X}$.
 - ④ Outer loop: Use (5) and estimate parameters via MLE until convergence.

What We Stuck

- Human capital investment
- Choice set may shrink
- Initial / termination condition

1. Introduction
2. Literature Review
3. Data
4. Estimation
5. Extension
6. Conclusion

Conclusion

- An upward trend in earnings is observed around the cutoff.
- The event-study estimates suggest that younger cohorts (born after 1994) earn more than older cohorts.
- However, the results lack robustness, partially due to data limitations.
- The use of repeated cross-sectional survey data poses challenges to identification.
- Multicollinearity between age, birth year, and survey year limits the feasibility of causal estimation.

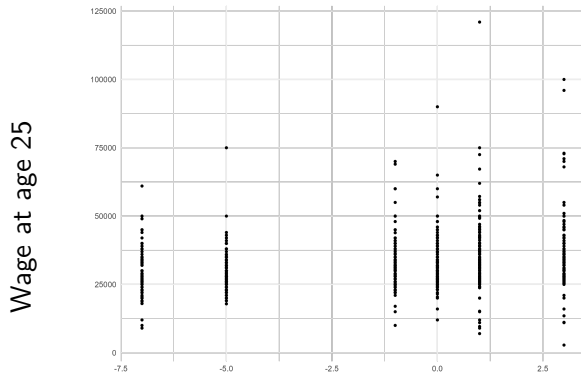
Table: Timeline of Military Service Reform

Year	Duration
1990	2 yr
2000	1 yr 10 mon
2004	1 yr 8 mon
2005	1 yr 6 mon
2006	1 yr 4 mon
2007	1 yr 2 mon
2008	1 yr
2013	4 mon
2024	1 yr reinstated

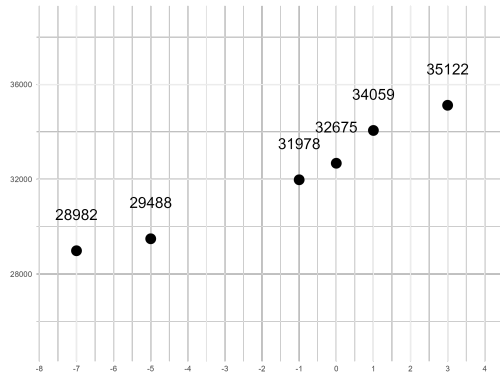
[Back](#)

Dependent Variable:	wage
<i>Variables</i>	
relative_birth_yr = -11	-2,690.5* (1,200.7)
relative_birth_yr = -10	-2,028.1 (1,119.4)
relative_birth_yr = -9	-2,585.7** (981.4)
relative_birth_yr = -8	-2,450.3** (785.1)
relative_birth_yr = -7	-1,793.1 (938.4)
relative_birth_yr = -6	-1,382.3* (700.3)
relative_birth_yr = -5	-1,276.2 (736.2)
relative_birth_yr = -4	-1,336.5* (625.2)
relative_birth_yr = -3	-521.0 (447.5)
relative_birth_yr = -2	-185.6 (446.1)
relative_birth_yr = 0	397.2** (140.7)
relative_birth_yr = 1	1,321.7*** (237.2)
relative_birth_yr = 2	776.9 (576.1)
relative_birth_yr = 3	1,472.5*** (338.5)
relative_birth_yr = 4	2,017.4* (938.3)
relative_birth_yr = 5	922.7 (609.4)
relative_birth_yr = 6	2,126.2** (606.5)
relative_birth_yr = 7	3,411.3** (1,015.8)
relative_birth_yr = 8	3,539.7** (1,400.0)
relative_birth_yr = 9	6,602.9*** (863.8)
bachelor	3,815.2*** (416.7)
yr	Yes
age	Yes
Observations	12,434
R ²	0.16817

Finite Mass Points



Raw Plot



Binned Scatter Plot

Relative Birth Year

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